PolymerFFKit User Manual

Operations, File formats, Illustrated guide, Software design

PolymerFFKit v1.0.0

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Before you begin

The program generates parameter estimates and initial structures. It does not run quantum chemistry, minimization, equilibration, or production dynamics. Check units, mapping, and connectivity before using outputs.

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01 Scope

This manual covers all six interactive steps in PolymerFFKit 1.0.0, their input and output formats, the Python interface, and the HTPB-cis example. It is intended for users extracting geometry distributions, fitting marginal potentials of mean force, and constructing initial polymer structures.

AA means all atom, UA united atom, and CG coarse grained. COM is the center of mass and COG the center of geometry. Both fitting methods operate of NLS (Nonlinear Least Squares) and MLE (Maximum Likelihood Estimation) on trajectory bond lengths, angles, and dihedrals.

The outputs are parameter estimates and initial structures. PolymerFFKit does not run quantum chemistry, minimization, equilibration, or production dynamics. Exported topology files require checking against the intended simulation engine.

02 Installation and startup

Windows x64: extract the entire ZIP to a writable directory and open the included launch script, or run PolymerFFKit.exe. The fitting launcher or PolymerFFKit.exe --fit opens Step 13 directly. Python is bundled. Allow time for runtime extraction on first launch.

Linux x86_64: this native package was built and tested in Fedora Linux 42 under WSL. After extracting the tar.gz, use the commands below. Consult BUILD_VERSIONS.txt and PLATFORM.txt. A GLIBC or GLIBCXX version error requires a compatible system or a new build on the target platform.

Keep the relative layout of data, templates, manual and example. Choose a writable output directory and allow additional space for trajectories and plots. Geometry observations accumulate in memory, so memory requirements grow with the number of frames and adjacency relationships.

For source use, run python -m pip install -r requirements.txt inside the PolymerFFKit directory, followed by python run.py. The syntax requires Python 3.10 or later, subject to the installed dependencies. The Windows build was tested with Python 3.12.3; the Linux build manifest records its actual environment.

```
tar -xzf PolymerFFKit-v1.0.0-linux-x86_64-fedora42.tar.gz
cd PolymerFFKit-v1.0.0-linux-x86_64-fedora42
chmod +x PolymerFFKit start.sh
./PolymerFFKit --version
./PolymerFFKit
./PolymerFFKit --fit
```

03 Menus and workflow

Main menu 1 opens trajectory analysis and force field derivation; 2 opens sequence and topology construction; 0 exits. Enter -1 in a submenu to return. Enter accepts a displayed default; Tab completes paths in an interactive terminal. Paths containing spaces may be quoted. Relative paths are resolved from the process working directory, not the input file directory.

11: original XYZ plus mapping JSON to mapped XYZ.

12: mapped XYZ plus distance cutoff to raw geometry TXT.

13: raw TXT or legacy histogram CSV plus fitting settings to parameter TXT, CSV and PNG.

14: mapping JSON, fitted TXT, ITP library and optional charges to unified JSON.

21: sequence settings to sequence TXT, composition TXT and configuration JSON.

22: sequence TXT, unified JSON and external ITP to PDB, ITP, TOP and DATA for each chain.

The two workflows connect through the unified configuration, but individual steps may be used separately.

Use separate output directories for different methods, temperatures, bin counts or models. Reusing a filename overwrites its contents. Ctrl+C interrupts processing, but partially written files can remain; check their completeness before reuse.

```
Main menu
1  Trajectory Analysis & Force Field Generation
   11  Mapping
   12  Raw geometry extraction
   13  NLS or MLE fitting
   14  Non-bonded parameter assignment
2  Polymer Sequence & Topology Builder
   21  Sequence generation
   22  Structure and topology export
0  Exit
```

04 XYZ trajectory format and units

Each frame starts with the atom count N, followed by a comment line and exactly N coordinate records. The comment may be empty, but its line must exist because the parser always consumes it. Concatenate frames directly without extra headers or blank separators. A coordinate record contains at least name, x, y and z, separated by whitespace, with decimal points in numeric values.

Input coordinates must be in angstroms. Step 11 preserves the coordinate unit. Step 12 multiplies distances by 0.1 to obtain nm and returns angles in degrees. Units, temperature, periodic cells and

lattice vectors are not inferred from comments. Writing T=500K in a comment does not set the fitting temperature.

Keep atom order and names consistent across frames so mapping indices remain valid. Make molecules whole before mapping. Calculations use ordinary Euclidean distances, without minimum image correction or automatic unwrapping. Check for zero length vectors, duplicated positions and incomplete frames.

Names become geometry labels. Use names such as LC1 and CR2 without spaces or embedded hyphens, because hyphens join interaction names. Interactions sharing the same label are pooled.

The example below contains two frames of three sites. It demonstrates syntax, not a sufficient fitting dataset.

```
3
Frame 0 coordinates in angstrom
LC1 0.000 0.000 0.000
CR2 2.300 0.000 0.000
LC3 3.600 1.800 0.000
3
Frame 1 coordinates in angstrom
LC1 0.000 0.000 0.000
CR2 2.320 0.010 0.000
LC3 3.610 1.790 0.010
```

05 Step 11 mapping and mapping rules

Select AA, UA or CG, enter the XYZ path, and specify the tail percentage to retain, greater than 0 and at most 100. A value of 80 keeps the last 80%; it is not periodic frame subsampling. AA only extracts frames. UA auto groups following hydrogen records with the preceding heavy atom based on file order, not chemical connectivity. Use UA custom when this ordering assumption is unsuitable.

CG and UA custom require mapping JSON. rules is an array of objects: name is the output bead name and atoms is a nonempty list of one-based indices into a frame. Indices refer to coordinate record order, not atomic numbers, and must lie between 1 and N. The user controls overlaps and omissions; unselected atoms are not added automatically.

The optional custom_masses dictionary maps exact XYZ names to masses in amu. COM is mass weighted; COG is an arithmetic mean. Without custom masses, the program infers elements from names and falls back to 1.0 when unrecognized. Supply explicit masses for nonstandard names.

Select 1 for COM or 2 for COG, then choose an output XYZ. Defaults are AA_extracted.xyz, UA_mapped.xyz or CG_mapped.xyz. Each output frame has one site per rule, named by name, with

coordinates still in angstroms. Automatically generated UA rules are not separately saved as mapping JSON.

JSON requires double quotes and permits neither comments nor trailing commas. The following example requires at least four atoms in the original trajectory.

```
{
  "custom_masses": {"C": 12.011, "H": 1.008},
  "rules": [
    {"name": "A", "atoms": [1, 2]},
    {"name": "B", "atoms": [3, 4]}
  ]
}
```

06 Step 12 geometry extraction

Enter the mapped XYZ, output raw file and BOND_CUTOFF in nm. Defaults are raw_analysis.txt and 0.8 nm. Create the output parent directory first. A cutoff of 0.8 nm is a default, not a universal CG chemical bond criterion; the 0.2 nm AA example also needs checking.

Pairs at or below the cutoff become neighbors in each frame. Bonds, angles and dihedrals are then enumerated from that adjacency. There is no fixed chemical topology input or distinction between intra- and intermolecular neighbors. Close nonbonded contacts may therefore appear as bonds. Select chemically meaningful labels before deriving a model.

Bond names are sorted, so LC1–CR2 is normally stored as CR2-LC1. Angle labels keep the central site second and sort the two endpoints. Dihedrals have an ordering and sign convention; do not relabel them arbitrarily. Samples with the same type are pooled across frames and interactions, without retaining frame identities.

07 Raw sample text and histogram CSV

Raw sample files use the header, optional comments and type records shown below. Store each label on one line, followed by a colon and comma separated finite numbers. Do not split a type across multiple lines. Bonds are in nm; angles and dihedrals in degrees. Three samples are only the minimum accepted input, not evidence of statistical adequacy or guaranteed convergence.

bond_cutoff_nm records the actual extraction upper bound used in MLE normalization. Legacy text may omit it; MLE then asks for the bound. Blank means no upper truncation. Do not substitute the observed maximum or an inferred range for the extraction threshold.

MLE accepts raw observations only. CSV binning is already fixed and cannot be changed during fitting. Extensions do not define content: HTPB_cis analysis.csv is a raw text dump and requires raw mode; histogram.csv requires csv mode.

```
Raw Analysis Data Dump (Grouped by Type)
# units: bond=nm, angle=degree, dihedral=degree
# bond_cutoff_nm: 0.8
BOND_A-B: 0.230,0.235,0.228,0.233
ANGLE_A-B-C: 109.0,110.0,111.0,109.5
DIHEDRAL_A-B-C-D: -170.0,175.0,179.0,-178.0

CSV example (save as a separate file)
center_bond,bond_A-B,center_angle,angle_A-B-C
0.220,10,100,12
0.230,50,110,48
0.240,11,120,13
```

08 Step 13 NLS operation

Choose nls, then raw or csv. Enter the input and output directory, default fit_results. Set bond and angle temperatures separately, default 300 K, and PMF thresholds, default 50%. Temperatures must be positive and match the trajectory used for fitting.

For dihedrals, 0 skips fitting, 1 selects Fourier and 2 selects RB. Fourier requires order 1–4. If enabled, specify the dihedral temperature and PMF threshold, default 100%. Choose whether to plot. Raw input always requests bins, default 60, an integer of at least 3. CSV input does not request bins.

NLS normalizes counts, computes the PMF and uses count weighted curve_fit regression. The threshold selects points by vertical PMF range: 50% retains values at or below the minimum plus half that range. It is neither half the frames nor a confidence interval. A low threshold or too few bins can leave too few fitting points.

For raw bond and angle data, histogram bounds follow the observed minimum and maximum. Dihedral histograms span -180 to 180 degrees. bins specifies a count, not a bin width. Assess sensitivity using identical samples, temperature, model and threshold, while saving separate results for counts such as 30, 60 and 100.

Invalid input or nonconvergence can cause an individual type to be skipped. Check each target success message and its CSV row. A normal process exit alone does not prove that all requested fits succeeded.

09 Step 13 MLE operation and models

Choose mle and enter raw sample text directly. CSV selection is unavailable. Confirm the bond extraction bound, choose the output directory, bond and angle temperatures, and the dihedral model. There is no PMF threshold prompt. With plotting enabled, bins controls display only; otherwise no bin count is requested.

MLE minimizes the negative log likelihood of exp(-U/kBT) divided by its normalization integral. Domains are zero to the extraction cutoff for bonds, or infinity without an upper bound; zero to 180 degrees for angles; and one full period for dihedrals. The normalizer depends on the parameters and cannot be omitted.

Bonds and angles use harmonic potentials. Fourier supports orders 1–4; RB is a cosine polynomial. Additive energy constants are unidentifiable in MLE and fixed to zero. Numerical quadrature nodes used for periodic integration are not histogram bins. Changing display bins does not alter estimated parameters.

Both methods retain the scalar coordinate marginal PMF convention, without dividing by r squared or sin(theta). The result is an effective marginal potential, not automatically an isolated bonded potential with geometric factors removed. Differences between methods can also arise from NLS thresholds and weights.

MLE removes bin dependence, but does not guarantee a correct potential family or remove temporal correlation. MLE parameter errors are NaN, meaning not estimated. NLS covariance errors are also not trajectory block uncertainties. Scientific inference requires separate block resampling retaining frame and interaction identity, or independent trajectories.

```
kB = 0.008314462618 kJ mol^-1 K^-1
U_harmonic(x) = 0.5 * k * (x - x0)^2 + c
U_Fourier(phi) = c + sum[k_i * (1 + cos(i*phi - d_i))]
U_RB(phi) = sum[C_i * cos(phi)^i], i = 0,...,5
p(x | parameters) = exp(-U(x)/kBT) / Z(parameters)
NLL = sum[U(x_j)/kBT] + N * log(Z)
```

10 Reading fitting outputs

forcefield_params.txt records the method and convention, followed by the Bonds & Angles table with Name, x0 and k. Display precision is four decimals for x0 and two for k. Step 14 imports this table, so its configuration uses rounded values. Use CSV for precise method comparisons.

forcefield_params.csv has one row per geometry type. name combines bond_, angle_ or dihedral_ with a label; type identifies the geometry; fit_method is nls or mle; temperature_K stores the temperature. x0 and k are the equilibrium coordinate and force constant; x0_err and k_err are error

fields. x0_unit and k_unit specify units: bond k is kJ/mol/nm squared; angle x0 is in degrees and angle k in kJ/mol/rad squared.

NLS records histogram_bins and pmf_threshold_pct, and n_samples for raw input. MLE includes n_samples, negative_log_likelihood, converged, message and uncertainty_method, with bond_upper_nm for bonds. MLE histogram_bins is blank/NaN; plot_histogram_bins appears when plotting. Other diagnostics may depend on the model. Blank does not mean zero.

Fourier parameters are P0=c, P1=k1, P2=d1, P3=k2, P4=d2, and so forth. Amplitudes are in kJ/mol and phases in degrees. RB uses C0 through C5 in kJ/mol. Retain dihedral_type and the potential definition. Check the RB angular origin against the downstream engine.

Each type can generate name_fit.png at 300 dpi. Blue points are histogram PMF values and the red line is the fitted potential. For MLE, the blue points are display only. Plot energy shifts do not change estimates. PNG files are neither raw observations nor independent validation or goodness of fit statistics.

```
=== Extracted Force Field Parameters ===
Fit method: MLE
--- Bonds & Angles ---
Name                        | x0 (nm/deg)     | k
bond_CR2-LC1                 | 0.2306          | 81025.63

CSV fields for a harmonic fit
name,type,fit_method,temperature_K,k,x0,k_err,x0_err,...
```

11 Step 14 nonbonded parameters and charges

Enter the mapping JSON, fitted TXT and output unified JSON, default ff_unified_config.json. Mapping name values define the set of sites to assign. Select AA, UA or CG, then choose a listed ITP, custom or skip. Library indices depend on the installation; do not assume option 1 always means Martini.

Libraries reside in data/ff_libraries/AA, UA and CG. custom asks for an ITP path; skip selects manual assignment. No quantum calculation or L-J optimization is performed. Optional charge TXT contains a name and charge per line, in elementary charge units. A repeated name replaces its earlier value.

For each site, enter the target FF type. If found, accept with yes or select customize to edit sigma in nm and epsilon in kJ/mol. Unknown types require manual values. Then confirm charge and mass in amu. The site name identifies the system site; type identifies a library entry, and they need not match. The importer reads the exact [atomtypes] section and interprets the final five fields as mass, charge, ptype, sigma and epsilon. The six-column example below is accepted. Avoid inline comments after

data. Complex macros, nested includes and C6/C12 parameter conventions are not automatically resolved by this simplified parser.

Unified JSON stores type, mass, charge and vdw_custom; manual L-J entries also contain sigma and epsilon. Step 22 does not automatically turn custom L-J values into a new library. Prepare the corresponding ITP separately. Step 14 imports bonds and angles from fitted TXT, not dihedral coefficients.

```
[ atomtypes ]
; name mass charge ptype sigma epsilon
TC4 45.0 0.0 A 0.47 3.50

Separate charges.txt
LC1 0.00
CR2 0.00
```

12 Unified configuration JSON fields

metadata may contain description, resolution and force_field_include. Step 22 still requires the actual ITP file; metadata alone does not locate it. atom_properties is keyed by site name. type, mass and charge determine the exported properties, with defaults used when absent.

bonded_parameters.bonds maps A-B to [r0_nm, k_kJ_per_mol_nm2], treating the pair as undirected. angles maps A-B-C, with B central, to [theta0_degree, k], accepting reversed order. The current exporter uses a cosine based angle function, so a fitted harmonic angle k is not automatically equivalent. A dihedrals field can retain records, but Step 22 does not read it to generate dihedrals.

defaults contains atom_type, mass, charge and bond. Missing bonds use the default pair. Missing angle parameters can result in a 114 degree placement angle but no exported angle interaction. Successful export therefore does not prove parameter completeness.

Without atom_naming, each sequence token becomes one site in a simple sequential chain. With atom_naming, a starting key is I + current + next unit, an internal key is previous + current + next, and an ending key is previous + current + F. Values are site name lists. I and F mark boundaries. A missing matching key skips that unit, so check the atom count.

branch_definitions contains branch_atom, parent_mapping with start/middle/end parent site names, and dihedral_offset_deg for initial placement. rigid_definitions maps unit names to extra_bonds pairs. It affects geometry construction, not engine rigid body constraints. The following bond-only teaching configuration is not a fitted HTPB-cis force field.

```
{
  "metadata": {"resolution": "CG"},
  "atom_properties": {
    "A": {"type": "TC4", "mass": 45.0, "charge": 0.0},
    "B": {"type": "TC4", "mass": 45.0, "charge": 0.0}
  },
  "bonded_parameters": {
    "bonds": {"A-B": [0.35, 1250.0]},
    "angles": {}, "dihedrals": {}
  },
  "defaults": {
    "atom_type": "TC4", "mass": 45.0,
    "charge": 0.0, "bond": [0.35, 1250.0]
  }
}
```

13 Step 21 sequences and all outputs

Open menu 2 then 21. Load JSON or enter unit names, matching ratios, degree of polymerization (DP), sequence count and strict mode. Ratios must be nonnegative with a positive total and match the unit list length. Use positive integers for DP and sequence count. strict=true assigns integer counts by the largest remainder method and shuffles them. false samples independently from the probabilities, so finite sequence compositions vary.

For reproducibility, specify seed in JSON. The default -1 uses system randomness; saving -1 does not reproduce the same sequence. Structure placement has additional random operations, so a sequence seed alone does not guarantee identical coordinates in another process.

The current generator and composition report operate on concatenated strings and character lengths. Prefer single character units such as A, B and C. Multicharacter names can make Length and percentages inconsistent with unit counts. Step 22 can tokenize explicitly separated multicharacter names, but that does not make Step 21 statistics token aware.

JSON fields are residue, ratios, degree of aggregation (including spaces), seed, number_of_sequences and strict as a JSON boolean. The example creates two length-10 sequences, each with six A and four B units.

An output prefix such as output/polymer creates polymer_sequences.txt, polymer_analysis.txt and polymer_config.json. Each sequence occupies a header starting >Sequence_ followed by one

complete sequence line. The analysis TXT is tab delimited with ID, Length and unit percentages. The JSON records the generation settings.

```
{
  "seed": 42,
  "number_of_sequences": 2,
  "strict": true,
  "residue": ["A", "B"],
  "ratios": [6, 4],
  "degree of aggregation": 10
}

Sequence file syntax
>Sequence_1|Len:4
ABAB

Analysis file syntax (tabs)
ID  Length  A       B
1   4       50.00%  50.00%
```

14 Step 22 structure construction

Inputs are sequence TXT, unified JSON, external nonbonded ITP, output directory (default topology_out), and the molecule count written to the GROMACS TOP (default 1). JSON and ITP are distinct inputs. The program attempts to swap reversed extensions, but follow the prompts in the intended order.

Sequence parsing recognizes a line starting >Sequence_ and its next line, or a record containing Content:. Arbitrary FASTA >ID headers are not guaranteed to work. Keep each sequence on one line. For multicharacter names, use explicit spaces as in LC1 CR2 LC3 CR4, or commas. Compact strings use longest known atom_properties names first.

Connectivity, bond lengths and angles define a free chain placed by BFS and NERF. Missing placement angles default to 114 degrees; torsion placement includes randomness. No excluded volume check, multichain packing or energy minimization is performed. The PDB is an unequilibrated single chain, not a prepared condensed phase system.

Each sequence generates CHAIN_1, CHAIN_2 and so forth, with .pdb, .itp, .top and .data outputs. The TOP molecule count changes only [molecules]; it does not duplicate PDB coordinates. Multiple sequences yield independent TOP files, not an automatically combined mixture.

The manual/format_examples directory contains a simple two-site chain for format and export testing, not a validated chemical model. The separate HTPB-cis example documents mapping and fitting of supplied trajectory data.

15 PDB ITP and TOP output formats

PDB: after REMARK, fixed width ATOM records contain serial, site name, residue name, chain, residue number, x/y/z, occupancy and temperature factor. Internal nm coordinates are converted to angstroms. CONECT uses one-based site indices; TER and END close the file. No periodic box is automatically assigned. Site names are truncated to four characters and residue names to three.

Molecular ITP: [moleculetype] gives the name and nrexcl=1. [atoms] contains nr, type, resnr, residue, atom, cgnr, charge and mass. [bonds] contains ai, aj, func=1, r0 and k. Matching angles produce [angles] with ai, aj, ak, func=2, theta0 and k. Indices are one based. No [dihedrals] section is generated automatically.

System TOP: includes the external force field basename, may append extracted [defaults], [atomtypes] and [nonbond_params], then includes CHAIN_N.itp and writes [system] and [molecules]. The external ITP is not copied automatically; make its include path valid.

If the external ITP already defines the same defaults or atom types, extracted TOP sections can duplicate them. Inspect the organization rather than blindly including CHAIN_N.top in another complete system TOP. Also verify nrexcl and 1–4 rules against the chosen force field.

The following teaching two-site chain illustrates the fields, not a production model.

```
[ moleculetype ]
CHAIN_1 1
[ atoms ]
1 TC4 1 A A 1 0.000 45.00
2 TC4 2 B B 2 0.000 45.00
[ bonds ]
1 2 1 0.3500 1250.0

TOP ending
[ system ]
PolymerSystem
[ molecules ]
CHAIN_1 1
```

16 LAMMPS DATA and potential conventions

The DATA header gives atoms, bonds, angles, dihedrals=0, impropers=0 and type counts, followed by xlo/xhi, ylo/yhi and zlo/zhi. Bounds extend 50 angstroms beyond coordinate extrema; they do not define a density equilibrated box. Masses contains type ID and mass. Atoms # full contains id, molecule-ID=1, type, charge and x/y/z. Bonds and Angles use integer type and site indices.

Lengths are angstroms and energies are converted to kcal/mol, consistent with units real. Bond Coeffs # harmonic uses K = k_GMX × 0.239006 / 100 / 2 and r0 = r0_nm × 10, including the harmonic factor convention. Angle Coeffs is labeled cosine/squared, with K = k_GMX × 0.239006 / 2 and equilibrium angles in degrees. Pair Coeffs and a complete simulation input script are not provided.

Important limitation: Step 13 fits harmonic angles, while Step 22 exports GROMACS func=2 and LAMMPS cosine/squared. These are different functions, and the code does not automatically convert curvature. Do not treat fitted angle constants as directly transferable. Match the intended function or refit and verify a conversion. The GROMACS manual gives the local relation k_cos $\sin^2(\text{theta0})$=k_harm; it should not be applied blindly near 0 or 180 degrees.

Step 14 does not import dihedrals, and Step 22 does not export dihedral/improper interactions, custom L-J overrides or rigid body constraints. Add and validate required terms in downstream engine inputs. The examples verify file generation, not a production GROMACS or LAMMPS simulation.

References are the official GROMACS bonded interaction and LAMMPS harmonic bond, cosine/squared angle and units documentation, linked in the appendix.

17 HTPB cis example layout and provenance

example_HTPB_cis/original preserves the supplied R1a-b/CCCC inputs, outputs, excluding __pycache__. original_manifest.json records relative paths, byte sizes and SHA-256 hashes.

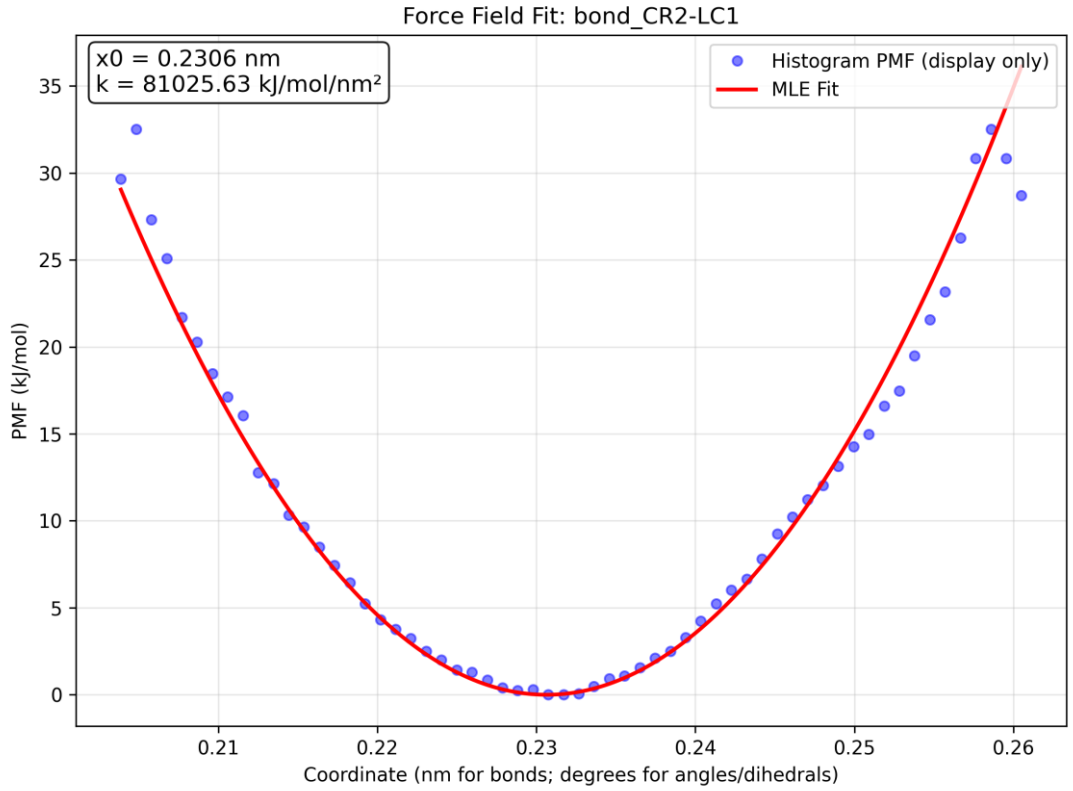
Main inputs are CCCC_trj.xyz and mapping_CCCC.json; existing mapped trajectories are CCCC_COM.xyz and CCCC_COG.xyz. analysis.csv is grouped raw sample text.

quickstart/inputs contains 500 consecutive AA frames starting at zero-based index 10000 and the mapping rules. Provenance was checked by matching computed COM coordinates to the first archived CCCC_COM.xyz frame. quickstart/expected contains current mapped XYZ, all raw geometries and target bond NLS/MLE results.

full_target/raw_CR2-LC1_90000.txt contains the 90000 target observations extracted from the original analysis.csv, with 0.8 nm metadata. full_target/expected stores current NLS/MLE outputs for 30, 60 and 100 bins.

The 500 K setting follows the earlier comparison; the XYZ header does not independently establish temperature, so verify the simulation record before scientific use. The consecutive 500-frame subset is for quick operation, not independent validation or convergence analysis.

```
example_HTPB_cis/
  original/                  archived supplied files
  original_manifest.json     original hashes
  provenance.json            extraction and fitting settings
  quickstart/
    inputs/                  500-frame AA and target raw files
    expected/                current software reference outputs
  full_target/
    raw_CR2-LC1_90000.txt
    expected/nls_30, nls_60, nls_100/
    expected/mle_30, mle_60, mle_100/
```



HTPB cis target bond MLE output 500 K fitting setting 90000 samples 60 display bins

18 HTPB cis mapping to fitting walkthrough

Create a rerun directory inside example_HTPB_cis/quickstart. In menu 1 → 11, select CG, inputs/HTPB_cis_AA_500frames.xyz, 100 percent, inputs/mapping_HTPB_cis.json, center 1 (COM), and output rerun/HTPB_cis_COM.xyz. Expect 500 frames with LC1, CR2, LC3 and CR4 in each frame.

In Step 12, use rerun/HTPB_cis_COM.xyz, output rerun/raw_all.txt and cutoff 0.8 nm. Expect six distance labels because the cutoff includes nonadjacent pairs among the four sites. The target is CR2-LC1; do not interpret all six types as chemical bonds automatically.

For target-only fitting, use the supplied inputs/raw_target_500frames.txt. In Step 13 choose nls, raw and rerun/nls_60; set both temperatures to 500 and thresholds to 50, skip dihedrals with 0, enable plots with y and set bins to 60. Repeat with mle and rerun/mle_60, temperatures 500, dihedrals 0, plots y and display bins 60. The bond-only input does not generate angles despite the angle prompt.

Repeat with 30 and 100 bins using separate directories and compare CSV values with expected. Estimates from the short subset can differ from the full dataset. Optimizers and numerical libraries can introduce small platform differences; plot files need not be byte identical.

For the full target, use full_target/raw_CR2-LC1_90000.txt with the same settings and a new rerun directory. With 90000 samples, NLS k is approximately 80909.09 at 30 bins and 80169.87 at 100 bins; MLE gives approximately 81025.63 kJ/mol/nm squared for both, with an equilibrium distance near 0.2306465 nm. This demonstrates bin behavior, not universal statistical superiority of MLE.

The complete archive and Linux program are packaged separately to avoid duplicating the large trajectories. The Linux package includes quickstart and full_target; extract the full example archive for original.

19 Python interface and batch comparison

Start Python from the parent of the PolymerFFKit directory so imports resolve. DataManager.load_raw_values returns dictionaries grouped by bond, angle and dihedral plus metadata. PMFAnalyzer accepts lower-case nls or mle for fit_method, default nls. bond_upper should come from the actual extraction metadata.

process_raw(values, name, pot_type, bins=60) returns a parameter dictionary and optionally saves PNGs when do_plot is enabled. It does not itself write the interactive TXT/CSV summary; callers should save returned values. pot_type is bond, angle or dihedral. Set params dihedral type to periodic or rb and Fourier n when needed.

The legacy process(bin_centers, frequencies, name, pot_type) interface remains available for NLS. MLE requires an additional raw_values argument; repeating bin centers does not reconstruct raw observations. examples/compare_fitting_methods.py supports comparisons across bin counts; use --help for its arguments.

For reproducibility, retain the data hash, software version, temperature, extraction cutoff, mapping mode, frame range, method, bin count, PMF threshold and model order. The following API

example prints bond results without writing a unified force field. Replace its input and temperature with actual settings.

```
from PolymerFFKit.core.traj_analysis import DataManager
from PolymerFFKit.core.ff_fitter import PMFAnalyzer

groups, meta = DataManager.load_raw_values("raw_analysis.txt")
fit = PMFAnalyzer("fit_mle", fit_method="mle",
                  bond_upper=meta.get("bond_cutoff_nm"))
fit.params["bond"]["T"] = 500.0
fit.do_plot = True
for label, values in groups["bond"].items():
    result = fit.process_raw(values, "bond_" + label,
                             "bond", bins=60)
    print(label, result)
```

20 Troubleshooting and result checks

File not found: check working directory, quotes and actual extensions. Choose raw or csv by content. JSON errors: remove comments and trailing commas and ensure an ITP was not supplied as JSON. Early trajectory termination: check atom counts, comment lines and blank separators.

MLE rejects a histogram: return to Step 12 and retain raw samples. Samples outside the normalization domain: check nm versus angstroms, angular units and extraction bounds. NaN/Inf, empty groups and constant values are unsuitable. Failed fits require checking sample adequacy, distribution and potential family, followed by NLS bins and threshold; more bins do not repair model mismatch.

Factors of ten or one hundred suggest unit or convention errors: check angstrom to nm, degrees to radians, kJ to kcal and harmonic factors of one half. MLE display points changing while parameters remain fixed is expected. Blank errors in CSV do not imply zero uncertainty.

Nonreproducible sequences: check whether seed is still -1. For misleading multicharacter composition statistics, use single character generation or prepare explicitly tokenized sequences. Missing topology atoms suggest incomplete atom_naming endpoint or internal keys. Missing angles suggest absent or mismatched interaction names.

Linux Permission denied: inspect execute permissions and noexec mounts; use a trusted executable location. GLIBC errors require checking platform compatibility. Use a UTF-8 terminal for display issues. Windows fitting with Chinese paths was tested.

Before scientific use, inspect target rows, convergence, units, sample counts and plots, then exported particle counts, total charge, types, interactions and engine functions. Validation includes numerical unit tests, source/executable fitting comparisons and example file generation, not comprehensive validation of every chemical system or simulation engine.

21 File index and references

File index: XYZ → 11/12; mapping JSON → 11/14; raw geometry TXT → 13 raw; histogram CSV → 13 nls csv; parameter TXT → 14; parameter CSV/PNG → comparison; external charges TXT → 14; force field ITP → 14/22; unified JSON → 22; sequence settings JSON → 21; sequence TXT → 22; composition TXT → reporting; PDB/ITP/TOP/DATA → inspection and engine preparation.

manual/format_examples provides copyable JSON, ITP, charge, sequence, XYZ and raw sample files, together with a teaching topology exported by the current program. Templates illustrate syntax, not recommended parameters. example_HTPB_cis supplies provenance, a quick reproduction path and full target fitting results.

Source responsibilities: run.py is the packaged entry point and __main__.py the main menu; function routes menus. core/mapping.py, traj_analysis.py, ff_fitter.py, mle_fitter.py, nonbonded.py, sequence.py and topology.py implement the main workflows. utils/io_utils/console.py handles terminal path input. The MLE module implements normalized likelihood; NLS remains in ff_fitter.py.

https://manual.gromacs.org/current/reference-manual/functions/bonded-interactions
https://docs.lammps.org/bond_harmonic.html
https://docs.lammps.org/angle_cosine_squared.html
https://docs.lammps.org/units.html

22 Additional output examples and diagnostics

The PDB and DATA excerpts below come from the two-site tutorial in format_examples/topology_expected. PDB retains fixed width columns; DATA uses whitespace with comments after #. Complete files accompany the manual, so there is no need to reconstruct spacing from PDF. Filename labels in the code block identify files and are not file content.

Additional MLE CSV fields: normalization_domain gives integration bounds; normalization_method identifies analytic or numerical normalization; optimizer_iterations counts iterations; gradient_norm gives a gradient norm; method identifies the estimator; density_measure records the coordinate density convention; energy_offset describes the energy origin; uncertainty

describes uncertainty treatment. Periodic models may include integration grid diagnostics. Read actual column names and inspect these together with converged and message.

negative_log_likelihood is the sum of negative log densities and can be negative without indicating an error. It depends on coordinate units, sample count and normalization domain. Values from different datasets, units or target types are not directly comparable rankings.

```
CHAIN_1.pdb
REMARK Free Floating Polymer Chain
ATOM      1 A    A   A   1       0.000   0.000   0.000  1.00  0.00           C
ATOM      2 B    B   A   2       3.500   0.000   0.000  1.00  0.00           B
CONECT    1    2
TER
END

CHAIN_1.data (selected sections; full file supplied)
2 atoms
1 bonds
0 angles
0 dihedrals
0 impropers
1 atom types
1 bond types
0 angle types
-50.000 53.500 xlo xhi
-50.000 50.000 ylo yhi
-50.000 50.000 zlo zhi
Masses
1 45.0000
Bond Coeffs # harmonic
1 1.4938 3.5000
Atoms # full
1 1 1 0.000000 0.0000 0.0000 0.0000
2 1 1 0.000000 3.5000 0.0000 0.0000
Bonds
1 1 1 2
```

23 Background and reading routes

PolymerFFKit is an interactive terminal tool for preparing polymer molecular dynamics models. The names the Department of Chemistry, Zhejiang University, and Tong Shi and Yingchun Liu. Distribution terms are defined by LICENSE.txt.

The software version remains PolymerFFKit-v1.0.0. The manual revision date, 2026-10-16, identifies this update; NLS/MLE selection, configurable bins and raw-sample output remain available.

New users should read installation, workflow and format chapters before running the HTPB cis quickstart. Users with existing parameters can start with sequences, unified configuration and topology construction. Developers can continue to architecture, algorithms, interfaces, performance and testing. The illustrated appendix contains all 17 original screenshots with instructions, output checks and current-version notes.

24 Hardware preparation and terminal interaction

The original documents suggest an x86-64 processor, 16 GB RAM, a 512 GB SSD and a 1920×1080 display. These are workstation suggestions, not measured minimum requirements. Storage needs depend on trajectories, figures, multiple output sets and extraction space. Memory depends on accumulated samples and adjacency relationships. A 512 GB drive recommendation does not mean the application occupies 512 GB.

Use PowerShell, CMD or Windows Terminal on Windows, and a UTF-8 terminal on Linux. Press Enter to accept the displayed default. Enter one of the listed choices. Tab completion is available in an interactive terminal; drag-and-drop path entry depends on the terminal. Quote paths with spaces. Relative paths are interpreted from the process working directory.

Main-menu choices 1 and 2 open parameter derivation and construction; 0 exits. Submenu choice -1 returns. Ctrl+C interrupts but does not roll back files already written. Use separate output folders and check nonempty files, target records and completion messages.

Processing AA input does not imply complete AA chemical model generation. Sequence and topology construction can run without a reference trajectory when external parameters are available, but the configuration and connectivity conventions must still match the builder.

25 Architecture and module responsibilities

The application separates interaction, function routing, core computation and shared utilities. run.py starts source or frozen execution and configures imports, UTF-8 streams and noninteractive plotting. __main__.py renders the banner, version and main menu. utils/io_utils/console.py provides

Rich console and path input. function/forcefield.py and function/builder.py dispatch menu choices and import core modules on demand.

Core responsibilities are mapping and masses in core/mapping.py; adjacency and geometry in core/traj_analysis.py; PMF/NLS and fitting interaction in core/ff_fitter.py; normalized likelihood estimation in core/mle_fitter.py; external parameter assembly in core/nonbonded.py; sequences in core/sequence.py; and coordinates and export in core/topology.py. Utility modules provide file helpers, although some writers are implemented directly in the core.

The design combines module-specific responsibilities, lazy imports, classes for data management and functions for calculations. JSON separates mappings, sequences and parameters from code. File-based interfaces allow external inputs and selective reruns, while requiring consistent labels and file versions.

```
run.py -> __main__.py -> main menu
  1 -> function/forcefield.py
       11 mapping.py -> 12 traj_analysis.py
       13 ff_fitter.py + mle_fitter.py -> 14 nonbonded.py
  2 -> function/builder.py
       21 sequence.py -> 22 topology.py
  Shared: utils/io_utils, JSON, XYZ, TXT, CSV, ITP
  Outputs: parameter records, plots, initial coordinates, topology
```

26 Complete data flow and handoffs

Parameter route: source XYZ and mapping definitions enter Step 11 and produce mapped XYZ. Step 12 takes mapped XYZ and a cutoff and writes raw geometry TXT. Step 13 performs NLS or MLE on raw observations; NLS also accepts a legacy histogram CSV. It writes parameter TXT/CSV and PNG plots. Step 14 takes mapping JSON, parameter TXT, external ITP and optional charges and writes unified JSON.

Construction route: Step 21 takes composition settings or sequence JSON and writes sequences, composition statistics and configuration. Step 22 reads sequence TXT, unified JSON and external ITP and writes per-chain PDB, ITP, TOP and DATA. It does not reread the source trajectory.

Check labels, units and file semantics at every handoff. Step 12 writes raw observations, not an automatic histogram.csv. Step 14 imports bond and angle values from the parameter TXT, not all CSV diagnostics or dihedral coefficients. Distribution fitting and simulation-system preparation require separate checks.

The original overall-flow, module-interface and fitting diagrams are consolidated below. The current flow explicitly includes MLE and raw input, resolving the older diagram label that described Step 12 output as CSV.

```
XYZ + mapping JSON -> 11 -> mapped XYZ
mapped XYZ + cutoff -> 12 -> raw geometry TXT
raw TXT -> 13 NLS -> histogram -> PMF -> fit -> TXT/CSV/PNG
raw TXT -> 13 MLE -> normalized likelihood -> TXT/CSV/PNG
legacy histogram CSV -> 13 NLS only
mapping JSON + fitted TXT + ITP + optional charges -> 14 -> unified JSON
sequence settings -> 21 -> sequences TXT + analysis TXT + config JSON
sequences TXT + unified JSON + ITP -> 22 -> PDB / ITP / TOP / DATA
```

27 Mapping and geometry algorithms

COM is $\Sigma m_i r_i / \Sigma m_i$, while COG is $\Sigma r_i / N$. Check element labels and custom_masses before COM mapping. Mapping JSON uses one-based atom indices, converted internally to zero-based indices. Trajectories are read frame by frame, but counting frames still scans the file. The processing percentage selects the trajectory tail rather than temporal subsampling.

Auto-UA groups by input order. A heavy atom starts a group, and following hydrogen records join that group. It does not infer hydrogen ownership from bond orders or distances. Heavy atoms and their attached hydrogens must therefore be adjacent; otherwise use custom mapping. AA mode retains coordinates and selects frames without adding missing hydrogens or assigning force-field types.

Geometry extraction builds an undirected distance graph per frame, converting angstroms to nanometres by multiplying by 0.1. Neighbour pairs define angles; consecutive four-site paths define dihedrals, with duplicates removed. Matching interaction labels pool their samples. A distance graph is not guaranteed to equal chemical connectivity, and large cutoffs include nonbonded neighbours.

Distances use Euclidean norms, angles use vector dot products and arccos, and signed dihedrals use plane normals and atan2. Make molecules whole and unwrap periodic coordinates before processing. The current workflow neither interprets cell metadata in XYZ comments nor applies minimum-image corrections.

28 Sequence allocation and coordinate construction

Strict mode normalizes composition weights, computes $q_i = DP \times w_i$, takes integer parts and assigns remaining positions to the largest fractional remainders. The monomer pool is then shuffled. For weights 5.5:0.5:4.0 and DP=11, raw counts are 6.05, 0.55 and 4.40, yielding 6, 1 and 4. Strict

mode preserves total DP and provides an integer approximation, not exact equality for every requested fraction.

Probabilistic mode samples each position independently using normalized weights, so individual compositions fluctuate. A fixed nonnegative seed reproduces sequences in the same environment. seed=-1 uses system initialization; saving that setting does not fix a sequence. The sequence seed also does not fix every random choice in coordinate construction. Weights must not all be zero. Single-character unit names avoid ambiguous string statistics.

The builder determines nodes from atom_naming and start, internal and terminal context keys. It applies backbone, branch and rigid_definitions connections, then uses breadth-first traversal to order placement. The first point is at the origin; placement with one reference follows the X axis. Three references allow local-frame placement.

NERF means Nearest-neighbor Extended Reference Frame. Let u follow $p_2 \rightarrow p_3$, n be the reference-plane normal and $v=n\times u$. With $\theta'=\pi-\theta$, $p_4=p_3+l\cos\theta' u+l\sin\theta'\cos\varphi v+l\sin\theta'\sin\varphi n$. Collinear references require fallback directions. This does not ensure a clash-free structure or an equilibrated ensemble.

Current initialization includes random backbone angles, branch angles drawn near dihedral_offset_deg with a 5-degree standard deviation, and 0/180-degree choices for rigid settings. These are not samples from fitted torsional probabilities and do not create engine rigid-body constraints. Inspect exported bonds, angles and coordinates.

```
sequence + compatible configuration
  -> resolve nodes and connections
  -> build adjacency and BFS traversal
  -> query bond lengths and angles
  -> choose reference points
       no reference: origin
       one reference: X direction
       two references: local plane
       three references: NERF placement
  -> inspect coordinates and graph
  -> export PDB / ITP / TOP / DATA
```

29 Runtime design and performance limits

Startup parses options such as --version and --fit, initializes the console and enters the menu loop. Function routing imports modules on demand; completed steps return to their submenu. File existence,

choice and numeric checks occur at different points, but not every error is recoverable. Read the first error and associated filename, retain the log and reproduce with minimal input.

Frame-wise mapping reduces memory use, but geometry analysis accumulates observations. Generator-based reading does not make all steps independent of trajectory size. Pairwise distance search scales as $O(N^2)$ in the number of sites; angle and dihedral enumeration also depends on graph degree. Dense neighbour graphs can be much more expensive than chains.

The original materials claimed 100–1000-fold speedups for vectorized distances without accompanying benchmark data. This manual does not present that figure as a performance guarantee. Speed depends on sites, frames, storage, cutoff and potential family. Periodic MLE adds integration and optimization costs. Confirm the 500-frame workflow before increasing data size.

Progress intervals differ between modules and are not success criteria. Output directories may be created automatically, while existing files can be overwritten. Failed fit types may be skipped. Check logs, file contents and target records rather than process exit alone.

30 Minimal construction tutorial and external simulation handoff

This tutorial complements the HTPB cis mapping and fitting example by illustrating construction with existing parameters. Use the sequence, unified JSON and external ITP in manual/format_examples. The two-site, one-bond teaching model illustrates file syntax; it is not an HTPB force field or a model for property prediction.

Select main menu 2 and Step 22. Supply the sequence, unified JSON and ITP in the documented order, choose a new output directory and set the molecule count to 1. Use the actual filenames listed in format_examples. Check two PDB coordinate records, one CONECT link and one bond in ITP/DATA. Compare with topology_expected before editing parameters. The molecule count changes TOP records; it does not pack additional coordinate copies.

To generate a sequence first, select Step 21 and load a configuration, or use A and B with equal weights, DP=2, strict mode and a fixed seed. Sequence labels must match the unified configuration. Longer sequences require corresponding connections and parameters.

Inspect the PDB with a molecular viewer. For GROMACS, check include paths, defaults, atomtypes and molecules and avoid duplicate definitions. Referenced ITP files must remain accessible. Do not indiscriminately include a complete TOP inside another complete TOP. For LAMMPS, choose consistent units and interaction styles before read_data and supply nonbonded settings. Both engines require further checking, minimization and equilibration; no unvalidated production input is provided here.

Current exports do not include complete dihedral terms. GROMACS angle functions and LAMMPS cosine/squared terms must be reconciled with the fitted potential. Do not assume that all custom Step 14 nonbonded parameters propagate into every export.

31 Source integration and correction index

The English screenshot guide contributes the introduction, two modules, environment notes, all 17 screenshots and their operating instructions. The Chinese operating manual contributes startup conventions, six steps, format tables, troubleshooting, the full workflow and template index. The design description contributes layering, module roles, data flows, interfaces, Auto-UA, adjacency, PMF, NERF, largest-remainder allocation and runtime design.

Figures in the two Chinese sources are primarily text diagrams, code examples or menu representations rather than embedded bitmaps. They are redrawn here as readable flow descriptions, equations and code blocks. All 17 actual bitmaps from the English source are retained unchanged, with historical interfaces distinguished from current operation.

Corrections replace older statements about general AA/UA/CG construction, cutoff-based chemical identification, automatic Step 12 CSV output, JSON // comments, exact arbitrary composition at finite DP, reproducibility with seed=-1, immediate simulation readiness, untested platform support and unmeasured speedup factors. NERF is consistently expanded as Nearest-neighbor Extended Reference Frame.

Original screenshot 15 reverses JSON and ITP paths; screenshot 16 shows the recovery. This manual states the correct order. Older fitting screenshots lack MLE and bin-count prompts; follow the current fitting chapters. Demonstration filenames, 300 K and bead types in those images are not parameter recommendations for HTPB cis.

Appendix A Illustrated operation guide

All 17 interface images from the original screenshot guide are retained. They document the original demonstration, not exclusively the HTPB cis data. Do not copy demonstration paths. Current source behavior and the main chapters take precedence over historical interface labels. Each image includes interpretation and checks.

A01 Startup and main menu

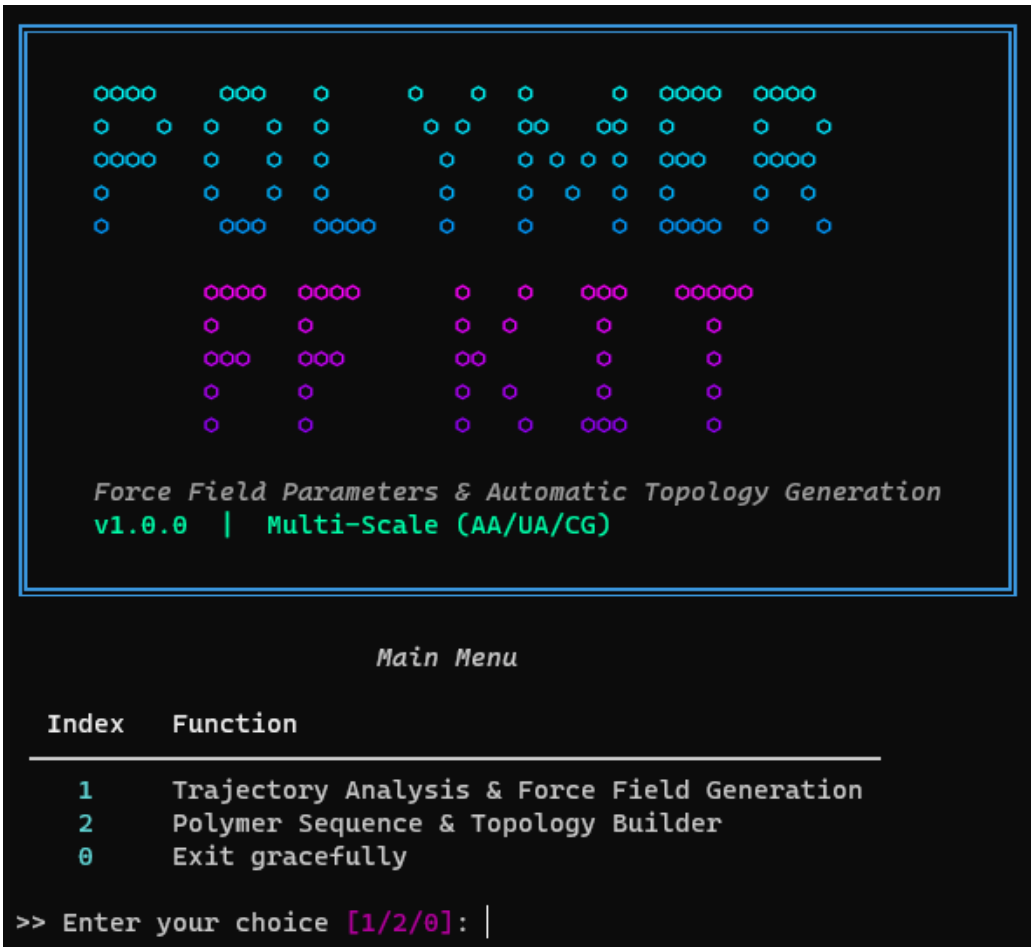


Figure A01 Original screenshot 1 Startup and main menu

Operation and verification

Start the program and check v1.0.0. Choose 1 for reference trajectories, 2 for construction from supplied configurations, or 0 to exit. Multi-Scale denotes processing modes, not validated arbitrary AA construction.

Use the corresponding main chapter to check input contents, units and output paths. Retain the run log and configuration. Demonstration settings in the image are not default physical parameters for your system.

A02 Parameter-route submenu

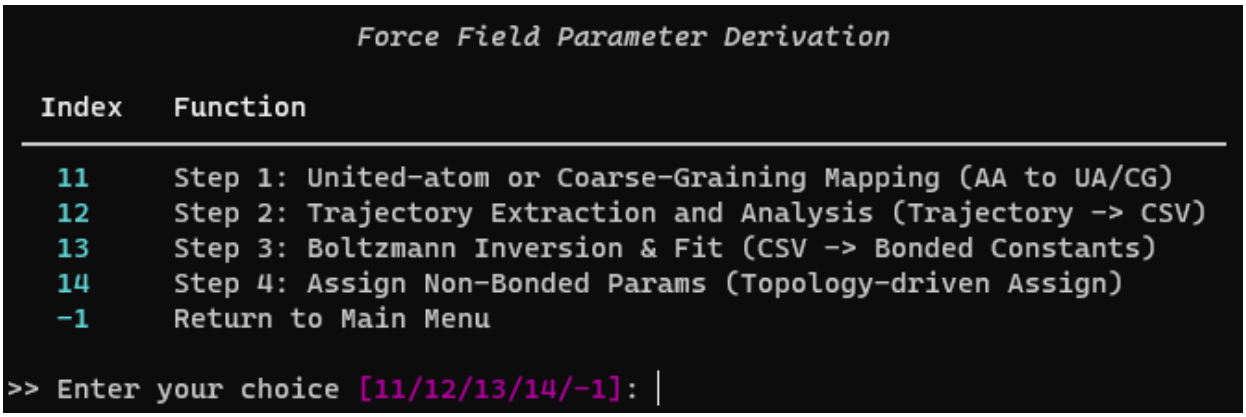


Figure A02 Original screenshot 2 Parameter-route submenu

Operation and verification

Follow 11, 12, 13 and 14 to understand file handoffs, or run a single step with suitable input. Trajectory to CSV is a historical label: current Step 12 writes raw TXT, and Step 13 supports raw NLS/MLE.

Use the corresponding main chapter to check input contents, units and output paths. Retain the run log and configuration. Demonstration settings in the image are not default physical parameters for your system.

A03 Mapping inputs

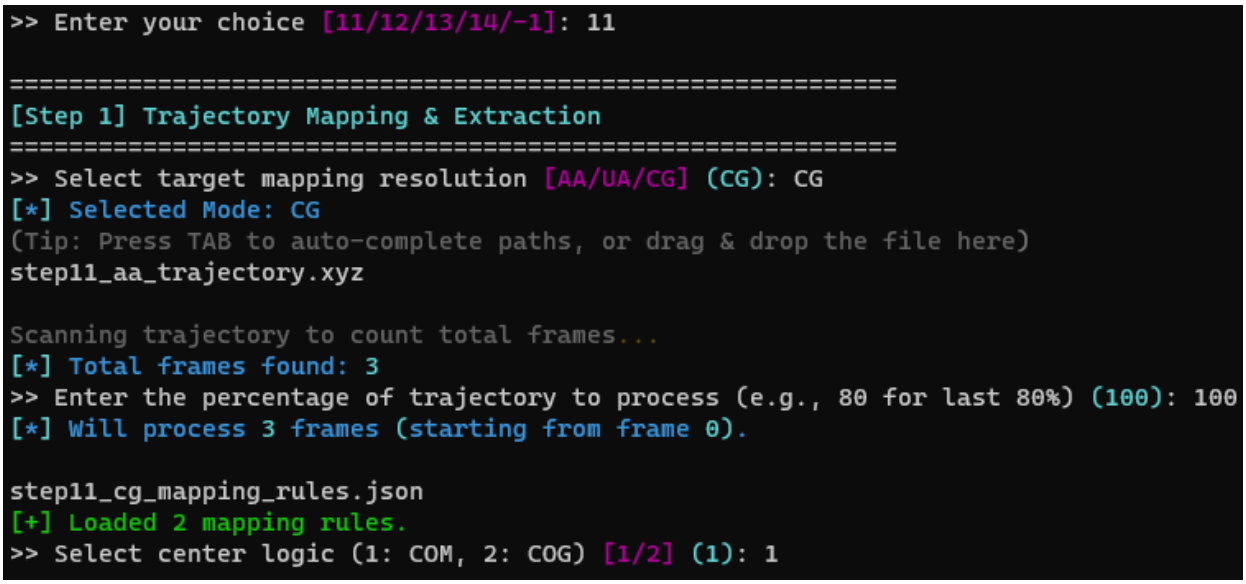


Figure A03 Original screenshot 3 Mapping inputs

Operation and verification

The image selects CG, all frames, two rules and COM. The HTPB cis quickstart uses 500 frames and four rules. Do not copy the three-frame demonstration counts or paths. Check total frames, start frame and rule count.

Use the corresponding main chapter to check input contents, units and output paths. Retain the run log and configuration. Demonstration settings in the image are not default physical parameters for your system.

A04 Mapping completion

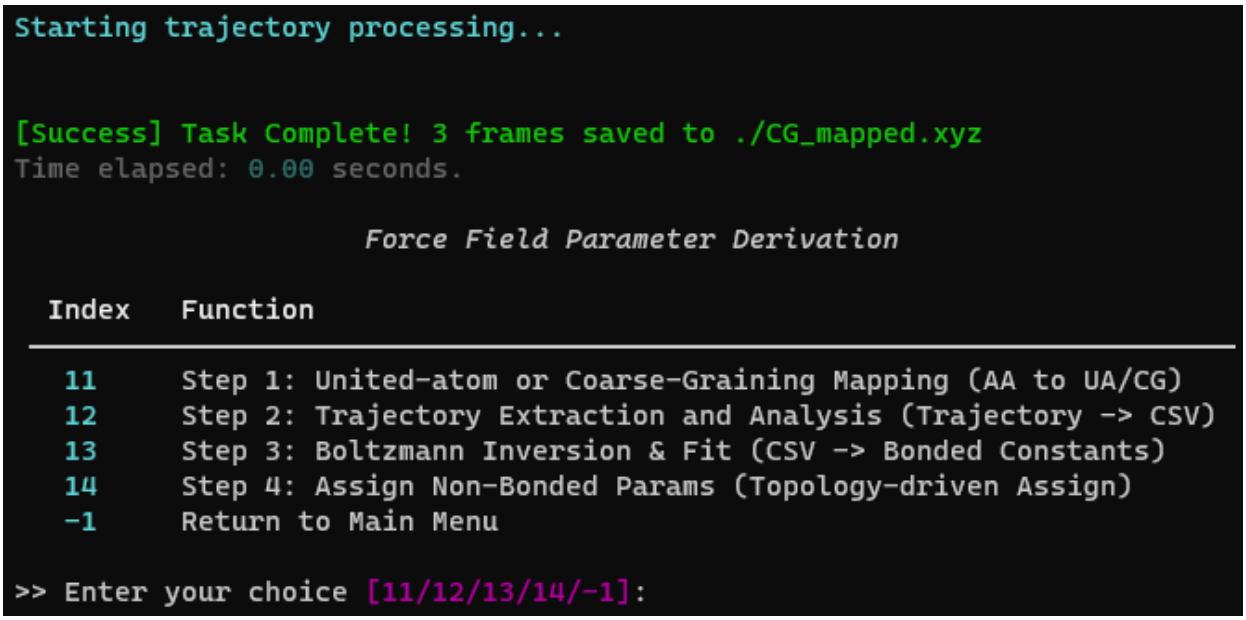


Figure A04 Original screenshot 4 Mapping completion

Operation and verification

Check the reported output path and frame count, then inspect site counts, labels and coordinate units in XYZ. Task Complete confirms writing, not chemical correctness of the mapping.

Use the corresponding main chapter to check input contents, units and output paths. Retain the run log and configuration. Demonstration settings in the image are not default physical parameters for your system.

A05 Geometry extraction settings

```
>> Enter your choice [11/12/13/14/-1]: 12

============================================================
[Step 2] Trajectory Geometry Extractor
============================================================
(Tip: Press TAB to auto-complete paths, or drag & drop the file here)
step12_cg_mapped_trajectory.xyz

>> Enter BOND_CUTOFF in nm (e.g., 0.8 for CG, 0.2 for AA) (0.8): 0.8

[*] Starting geometry extraction with Cutoff = 0.8 nm...

[*] Finished processing 3 frames.
```

Figure A05 Original screenshot 5 Geometry extraction settings

Operation and verification

Enter mapped XYZ, the raw-output path and cutoff in nm. The 0.8 nm value is an example setting, not a universal CG bonding rule. Unwrap periodic molecules first.

Use the corresponding main chapter to check input contents, units and output paths. Retain the run log and configuration. Demonstration settings in the image are not default physical parameters for your system.

A06 Geometry extraction results

```
--- Identified Geometry Types ---
  [+] Bonds: 4 types (e.g., ['Nda-TC4', 'TC4-TC4', 'P4-TC4']...)
  [+] Angles: 8 types (e.g., ['Nda-TC4-TC4', 'Nda-TC4-P4', 'P4-TC4-TC4']...)
  [+] Dihedrals: 20 types
[+] Raw analysis data saved to ./raw_analysis.txt

[Success] Geometry extraction completed successfully.
Time elapsed: 0.01 seconds.

                    Force Field Parameter Derivation

  Index    Function
 ------------------------------------------------------------------------
   11      Step 1: United-atom or Coarse-Graining Mapping (AA to UA/CG)
   12      Step 2: Trajectory Extraction and Analysis (Trajectory -> CSV)
   13      Step 3: Boltzmann Inversion & Fit (CSV -> Bonded Constants)
   14      Step 4: Assign Non-Bonded Params (Topology-driven Assign)
   -1      Return to Main Menu

>> Enter your choice [11/12/13/14/-1]:
```

Figure A06 Original screenshot 6 Geometry extraction results

Operation and verification

Bonds, Angles and Dihedrals counts refer to label types, not frames or total observations. Inspect labels and sample counts in the raw file. A distance graph can include nonchemical neighbours.

Use the corresponding main chapter to check input contents, units and output paths. Retain the run log and configuration. Demonstration settings in the image are not default physical parameters for your system.

A07 Original NLS interface

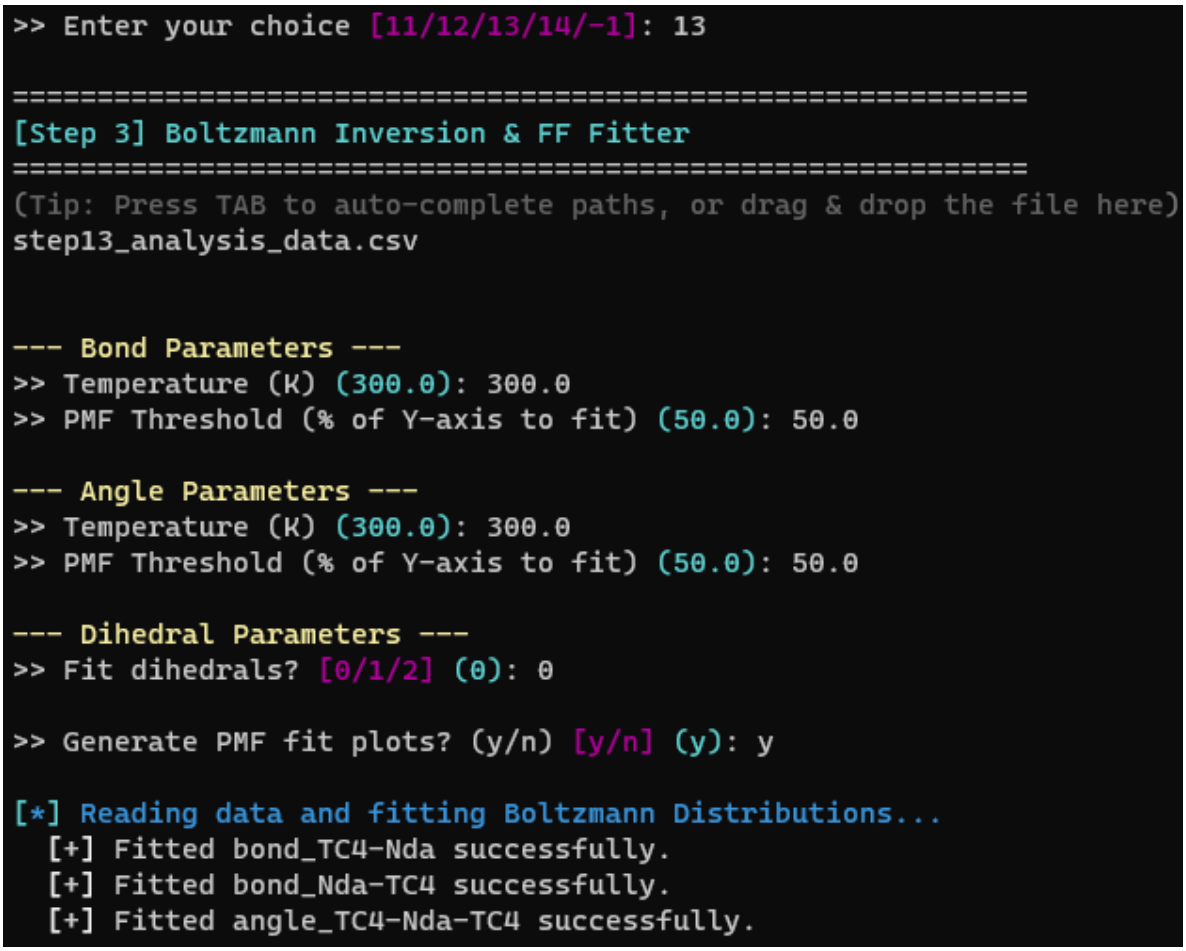


Figure A07 Original screenshot 7 Original NLS interface

Operation and verification

This is the original NLS path: temperatures, PMF thresholds, dihedral model and plotting. Current operation first selects nls or mle; NLS then chooses raw/csv. Raw bins are configurable, and MLE has no PMF-threshold prompt.

Use the corresponding main chapter to check input contents, units and output paths. Retain the run log and configuration. Demonstration settings in the image are not default physical parameters for your system.

A08 Fitting output checks

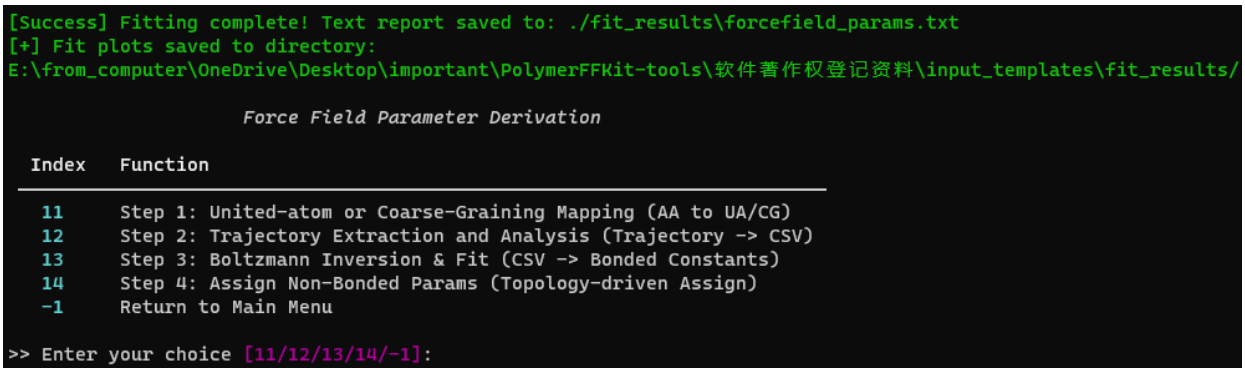


Figure A08 Original screenshot 8 Fitting output checks

Operation and verification

Check success for every target and its CSV row. Current structured CSV records method, temperature, sample count and diagnostics. Use the text report for assembly and CSV for precision comparisons.

Use the corresponding main chapter to check input contents, units and output paths. Retain the run log and configuration. Demonstration settings in the image are not default physical parameters for your system.

A09 Nonbonded library and charge inputs

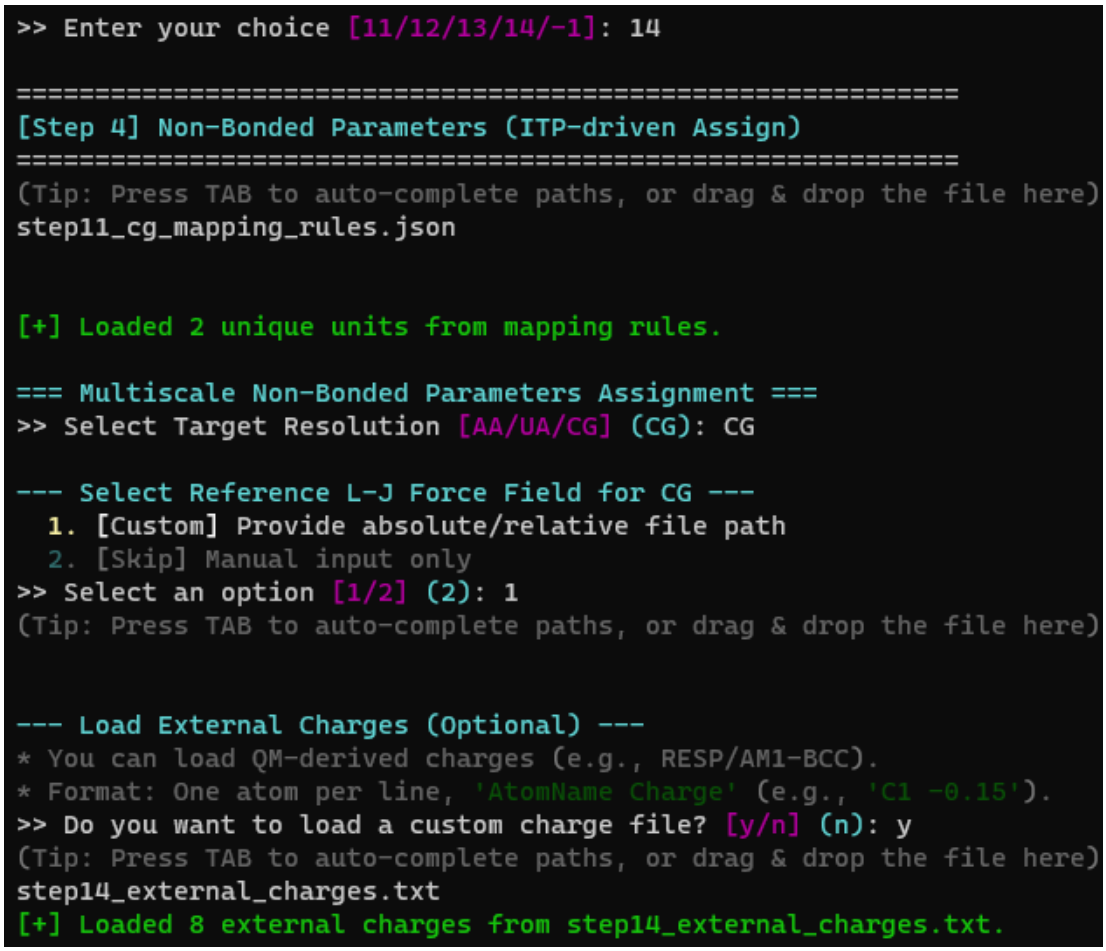


Figure A09 Original screenshot 9 Nonbonded library and charge inputs

Operation and verification

Supply mapping JSON and fitted TXT, select a library or manual parameters, and optionally load charges. The image lists custom/skip options; built-in choices depend on data resources. Charge keys must match mapped site names.

Use the corresponding main chapter to check input contents, units and output paths. Retain the run log and configuration. Demonstration settings in the image are not default physical parameters for your system.

A10 Per-site parameter assignment

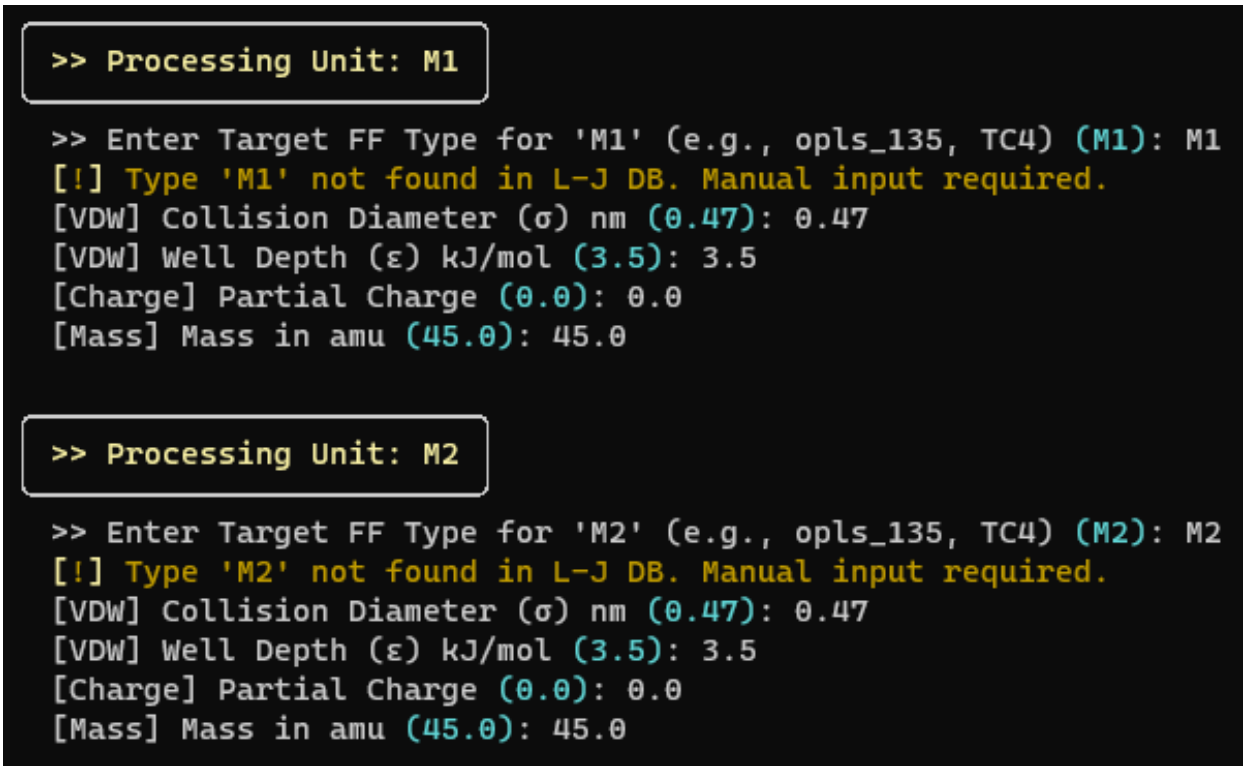


Figure A10 Original screenshot 10 Per-site parameter assignment

Operation and verification

Missing types require manual sigma, epsilon, charge and mass. Units are nm, kJ/mol, elementary charge and amu. The displayed 0.47, 3.5 and 45.0 values are examples, not universal bead recommendations.

Use the corresponding main chapter to check input contents, units and output paths. Retain the run log and configuration. Demonstration settings in the image are not default physical parameters for your system.

A11 Saving unified configuration

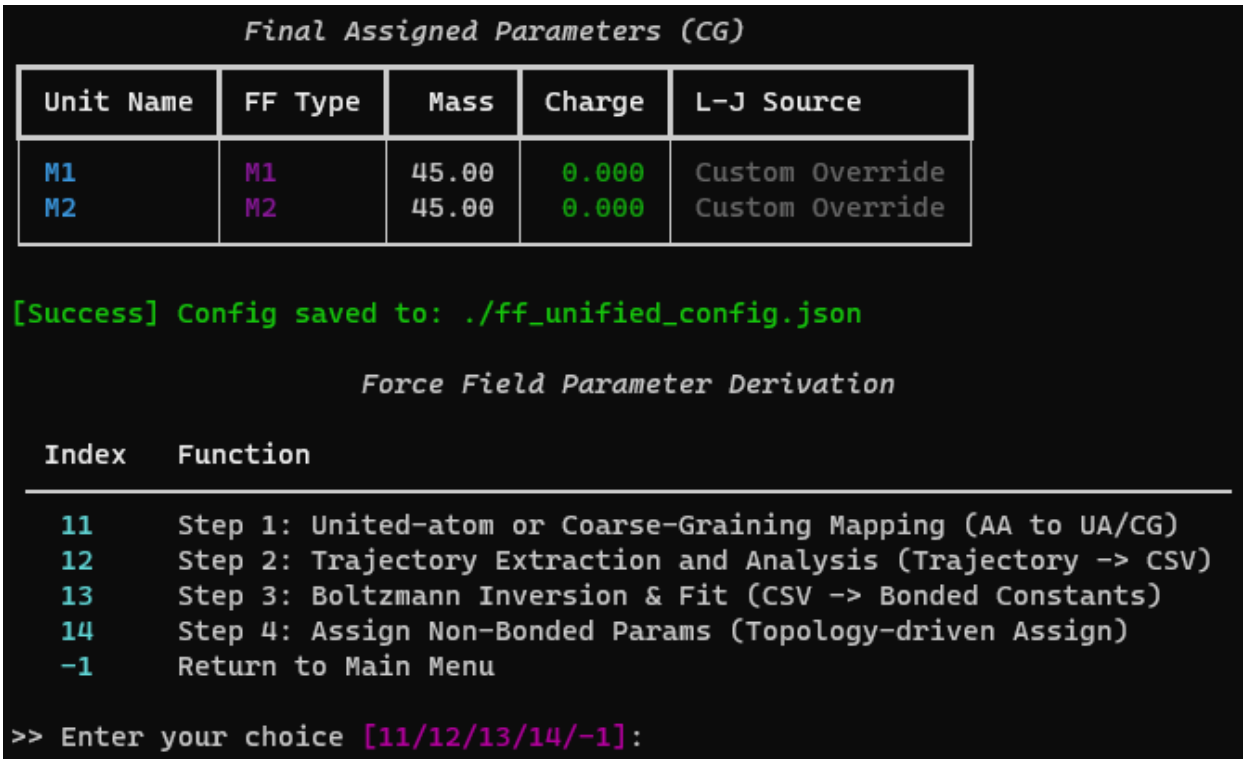


Figure A11 Original screenshot 11 Saving unified configuration

Operation and verification

Review Unit Name, FF Type, Mass, Charge and L-J Source, then inspect the saved JSON. Check bond/angle coverage for the sequence. Saving does not establish consistent exported potentials or complete nonbonded overrides.

Use the corresponding main chapter to check input contents, units and output paths. Retain the run log and configuration. Demonstration settings in the image are not default physical parameters for your system.

A12 Construction-route submenu

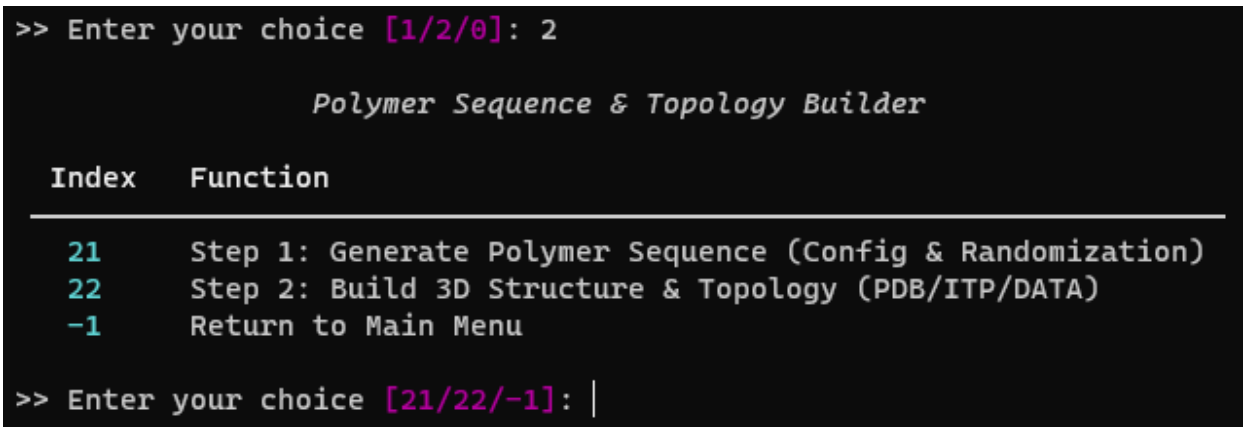


Figure A12 Original screenshot 12 Construction-route submenu

Operation and verification

Step 21 generates sequences; Step 22 constructs coordinates and topology from sequences and parameters. A valid existing sequence can enter Step 22 directly. Omitting a reference trajectory does not remove configuration requirements.

Use the corresponding main chapter to check input contents, units and output paths. Retain the run log and configuration. Demonstration settings in the image are not default physical parameters for your system.

A13 Loading sequence settings

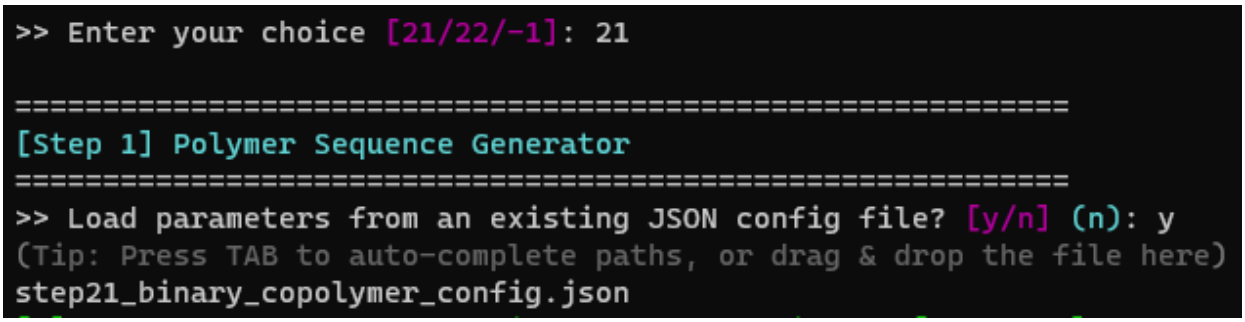


Figure A13 Original screenshot 13 Loading sequence settings

Operation and verification

The image loads an existing JSON. Manual input requires units, weights, DP, count and strict mode. Use a fixed seed for reproduction and preserve exact keys such as residue and degree of aggregation.

Use the corresponding main chapter to check input contents, units and output paths. Retain the run log and configuration. Demonstration settings in the image are not default physical parameters for your system.

A14 Three sequence outputs

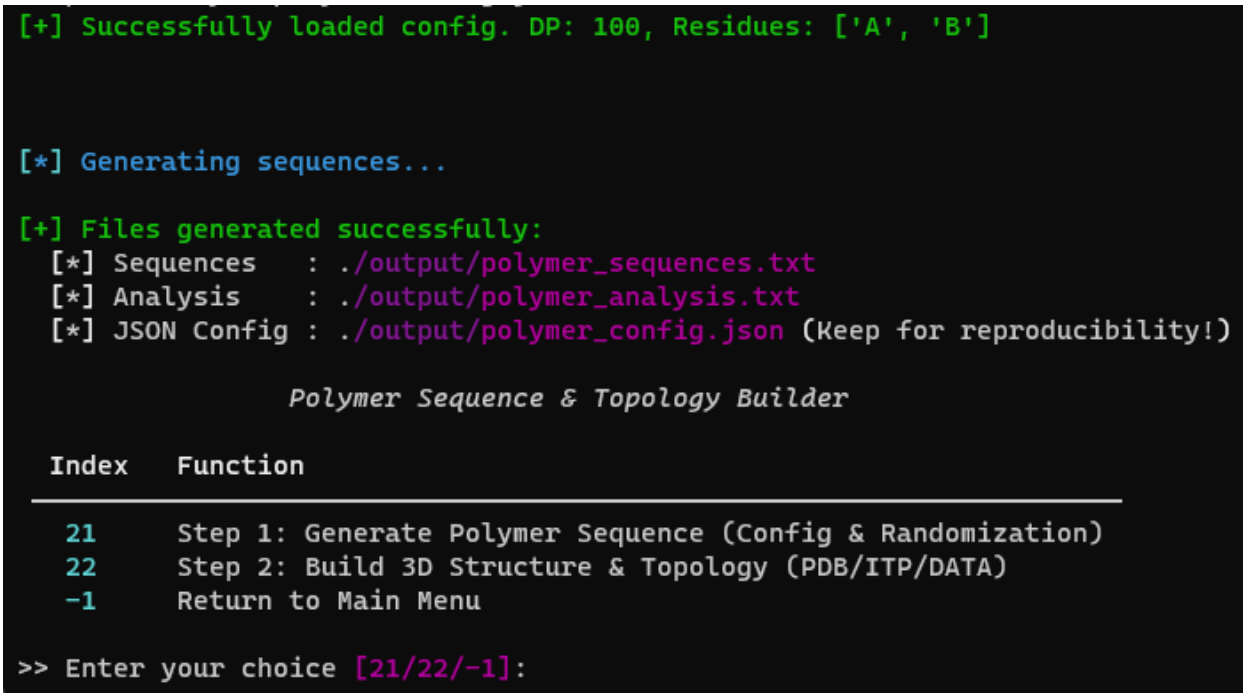


Figure A14 Original screenshot 14 Three sequence outputs

Operation and verification

Keep _sequences.txt, _analysis.txt and _config.json. Verify lengths and integer composition. Multicharacter labels affect current string-based statistics; the teaching example uses A/B.

Use the corresponding main chapter to check input contents, units and output paths. Retain the run log and configuration. Demonstration settings in the image are not default physical parameters for your system.

A15 Structure input order

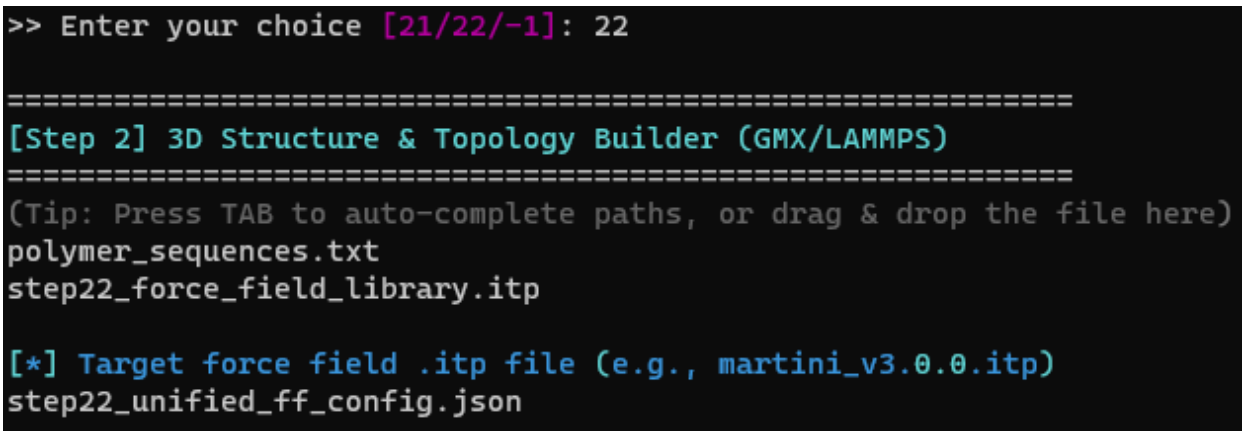


Figure A15 Original screenshot 15 Structure input order

Operation and verification

The original image reverses ITP and JSON inputs and is retained to explain recovery. Correct order is sequence TXT, unified JSON, external ITP, output directory and molecule count. Do not rely on automatic swapping instead of checking inputs.

Use the corresponding main chapter to check input contents, units and output paths. Retain the run log and configuration. Demonstration settings in the image are not default physical parameters for your system.

A16 Construction completion and limits

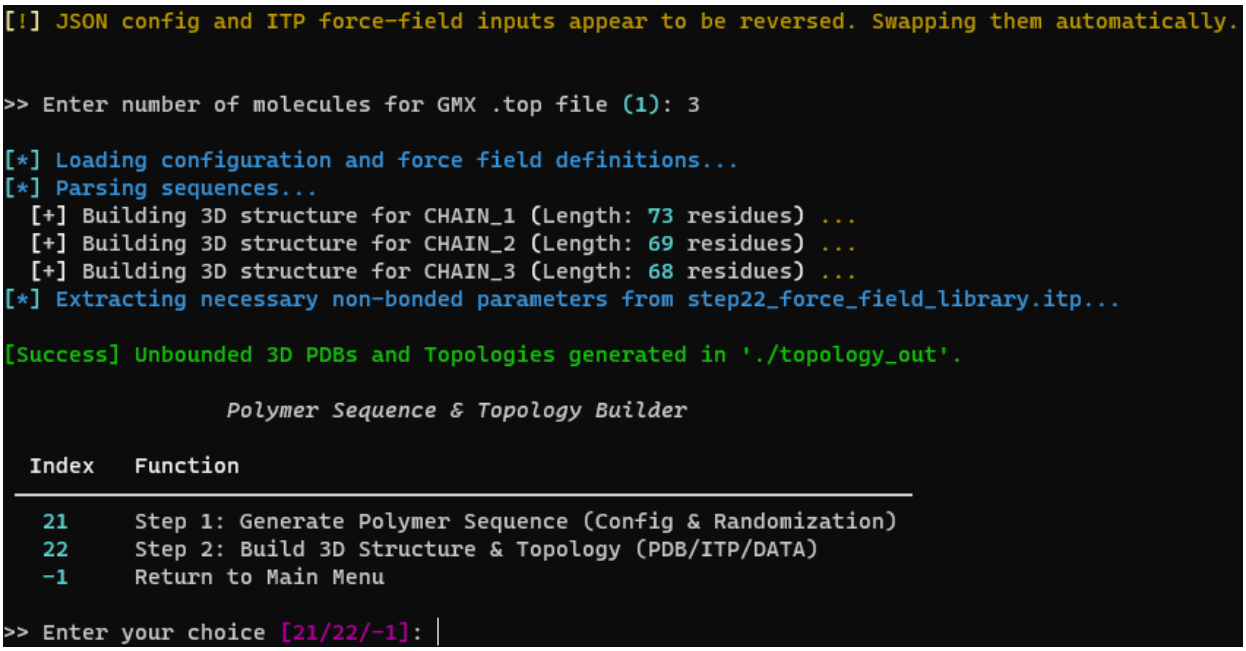


Figure A16 Original screenshot 16 Construction completion and limits

Operation and verification

The image shows input-swap recovery and multiple chains. Check generated chain counts, site counts and connectivity. TOP molecule counts do not duplicate or pack coordinates. Inspect PDB/ITP/TOP/DATA before external simulation preparation.

Use the corresponding main chapter to check input contents, units and output paths. Retain the run log and configuration. Demonstration settings in the image are not default physical parameters for your system.

A17 Normal exit and interruption

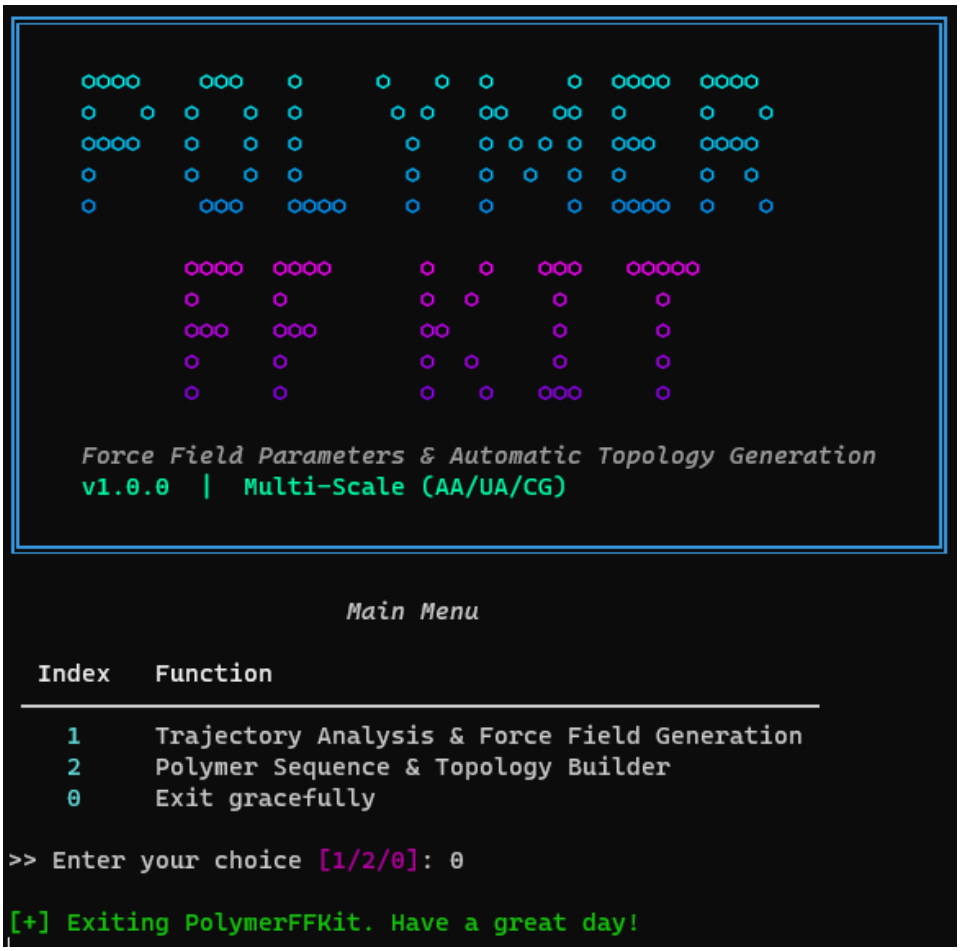


Figure A17 Original screenshot 17 Normal exit and interruption

Operation and verification

Enter 0 at the main menu to exit; return from submenus using -1 first. Ctrl+C interrupts, but existing files may be incomplete. Rerun in a new directory and verify outputs again.

Use the corresponding main chapter to check input contents, units and output paths. Retain the run log and configuration. Demonstration settings in the image are not default physical parameters for your system.

Appendix B File index

The following files accompany this manual in format_examples. Copy inputs into a working directory and keep expected outputs separate from reruns.

charges.txt

forcefield_minimal.itp

four_atoms.xyz

histogram_template.csv

mapping_minimal.json

raw_samples_demo_300K.txt

README.txt

sequence_AB.txt

sequence_demo_analysis.txt

sequence_demo_config.json

sequence_demo_sequences.txt

topology_expected/CHAIN_1.data

topology_expected/CHAIN_1.itp

topology_expected/CHAIN_1.pdb

topology_expected/CHAIN_1.top

unified_minimal.json